AVITI (Short-read)

Raw FASTQ

Paired-end
~12.7 M reads

Demux

cutadapt, e=0.1, overlap ≥15 bp
IGH / IGK / IGL primers
(run independently per chain)

IGH / IGK / IGL — run independently, same pipeline per chain

R1 Trim

cutadapt, e=0.1, overlap ≥15 bp
TSO trim only (no Q/length filter)
validated: 100% of demux reads retained

R2 Trim

cutadapt, e=0.1, overlap ≥10 bp
Q≥20, 150–300 bp
(100% of demux reads, no merge)

IgBLAST (R1)

igblastn, -outfmt 19
rabbit germline DB
-extend_align5end / 3end
(stop-codon check only; junction_aa ~0.2%)

IgBLAST (R2)

igblastn, -outfmt 19
rabbit germline DB
-extend_align5end / 3end
(main source: V/D/J, junction_aa, productive)

Stitch + QC

germline-coordinate splice (same read ID, not a physical merge)
exclude R1/R2 coverage-gap reads (~7–13%)
Biopython local alignment: overlap identity ≥90% (FLASH2-style QC),
exclude unreliable overlaps

IgBLAST (Stitched)

igblastn, -outfmt 19
rabbit germline DB
-extend_align5end / 3end
(full V(D)J: V/D/J, junction_aa, productive, v_identity — 3rd IgBLAST pass)

Nanopore (Long-read)

Raw FASTQ

Long reads
~5,400 reads

Filter

NanoFilt, Q≥10
350–600 bp (IGH)
320–600 bp (IGK/IGL)

Trim

cutadapt, e=0.1
overlap ≥15 bp
--revcomp, --discard-untrimmed

IgBLAST

igblastn, -outfmt 19
rabbit germline DB
-extend_align5end / 3end

AIRR TSV → BCR Repertoire Analysis

Productive reads | V/D/J assignment | junction_aa
Same productive definition & IgBLAST params across AVITI (IGH/IGK/IGL) and Nanopore

Analysis-condition equivalence: IgBLAST parameters (-outfmt 19, rabbit germline DB, -extend_align5end/3end, same internal_data) and the productive-read definition (productive field, or the stop_codon=F & vj_in_frame=T fallback) are identical across the AVITI Stitched lane (3rd IgBLAST pass, run independently for IGH/IGK/IGL) and Nanopore; only demux primers and chain-specific IgBLAST retry depth differ by locus.